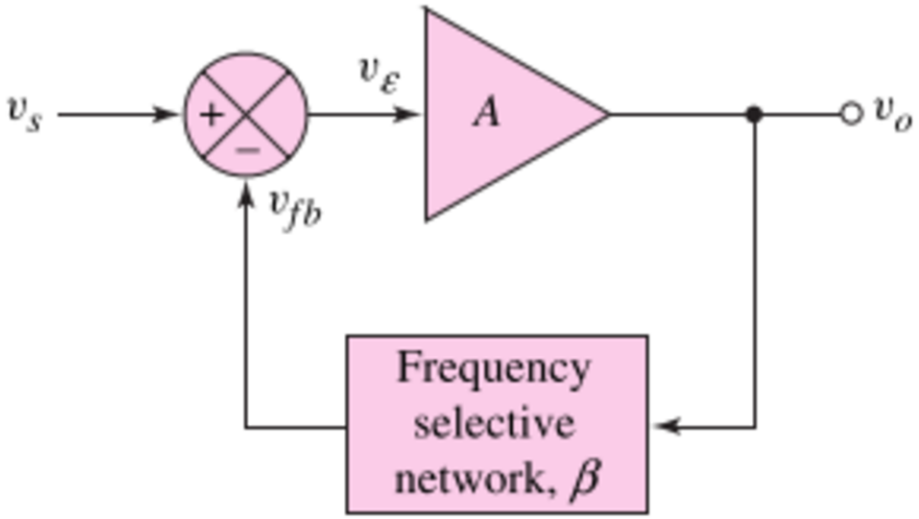
Oscillator

- Any kind of radio-TV transmitter or receiver or any laboratory test equipment has the oscillator.
- **It is the main component for producing the clock signal**.
- A simple oscillator application can be seen inside a very common device such as a watch.
- Watches use an oscillator to produce a 1 Hz clock signal.

LC circuit

- https://en.wikipedia.org/wiki/File:Tuned_circuit_animation_3_300ms.gif



Ideal closed loop transfer function

$$A_f(s) = \frac{A(s)}{1 + A(s)\beta(s)}$$

Loop gain of feedback circuit is

$$T(s) = A(s)\beta(s)$$

Loop gain T(s) is positive for negative feedback, which means feedback signal subtracts from input signal Vs.

If loop gain T(s) becomes negative, then feedback signal phase causes Vfb, to add to input signal, increasing error signal V_ε.

If T(s) = -1, closed loop transfer function goes to infinity, which means circuit can have finite output for zero input signal.

As $T(s)$ approaches -1, an actual circuit becomes nonlinear, which means that the gain does not go to infinity. Assume that $T(s) \approx -1$ so that positive feedback exists over a particular frequency range. If a spontaneous signal (due to noise) is created at v_s in this frequency range, the resulting feedback signal v_{fb} is in phase with v_s, and the error signal v_ε is reinforced and increased. This reinforcement process continues at only those frequencies for which the total phase shift around the feedback loop is zero. Therefore, the condition for oscillation is that, at a specific frequency, we have

$$T(j\omega_o) = A(j\omega_o)\beta(j\omega_o) = -1$$

The condition that $T(j\omega_o) = -1$ is called the **Barkhausen criterion.**

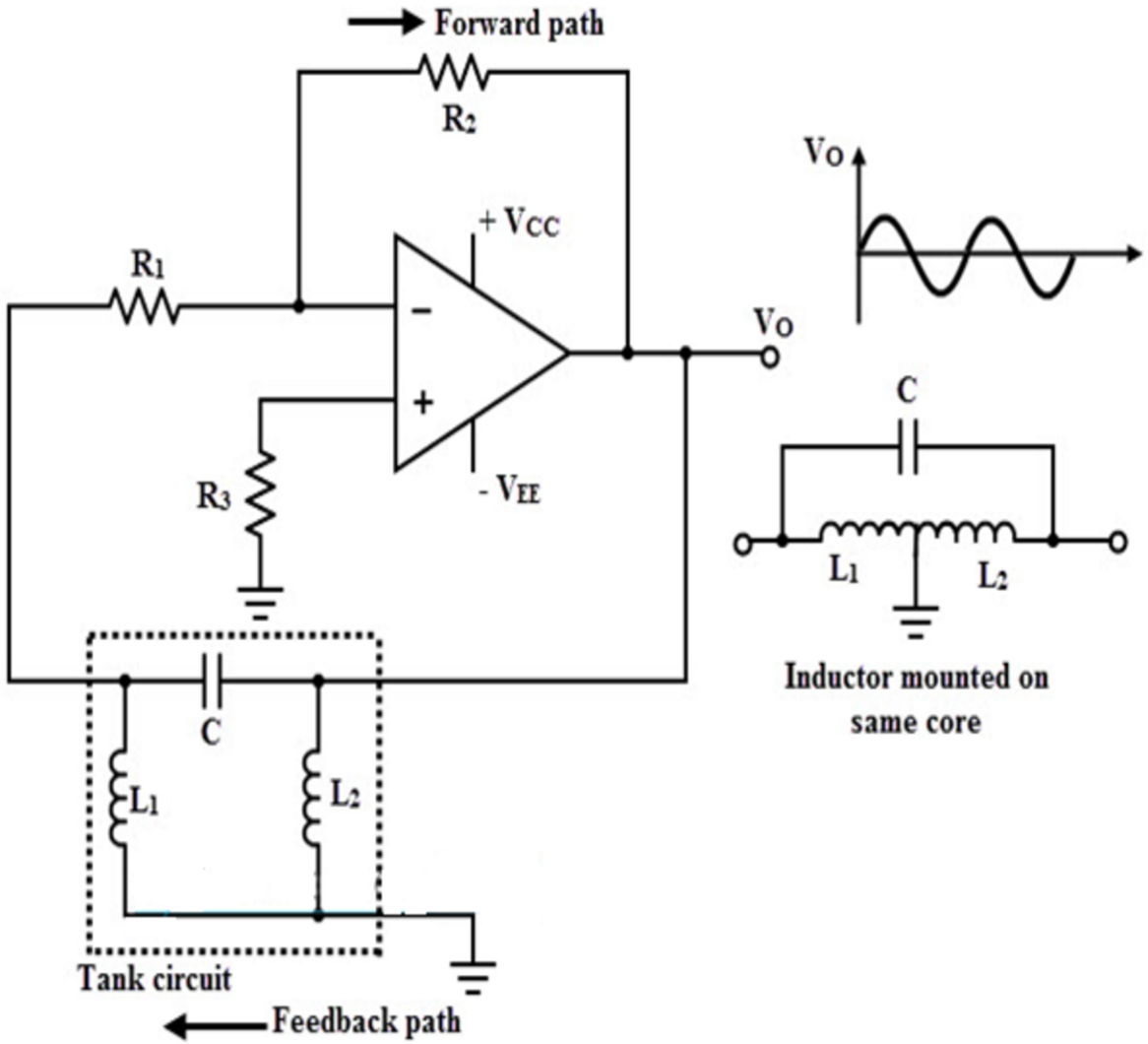
Two conditions must satisfy to sustain oscillation

1. Total phase shift through amplifier and feedback network must be N * 360
   N= 0, 1, 2, 3……
2. Magnitude of loop gain must be unity

- For an Oscillator feedback transfer function, or frequency selective network, must introduce additional 180° phase shift such that net phase shift around the loop is zero.
- For the circuit to oscillate at single frequency ωo, condition for oscillation, from following equation should be satisfied only at one frequency.

$$T(j\omega_o) = A(j\omega_o)\beta(j\omega_o) = -1$$

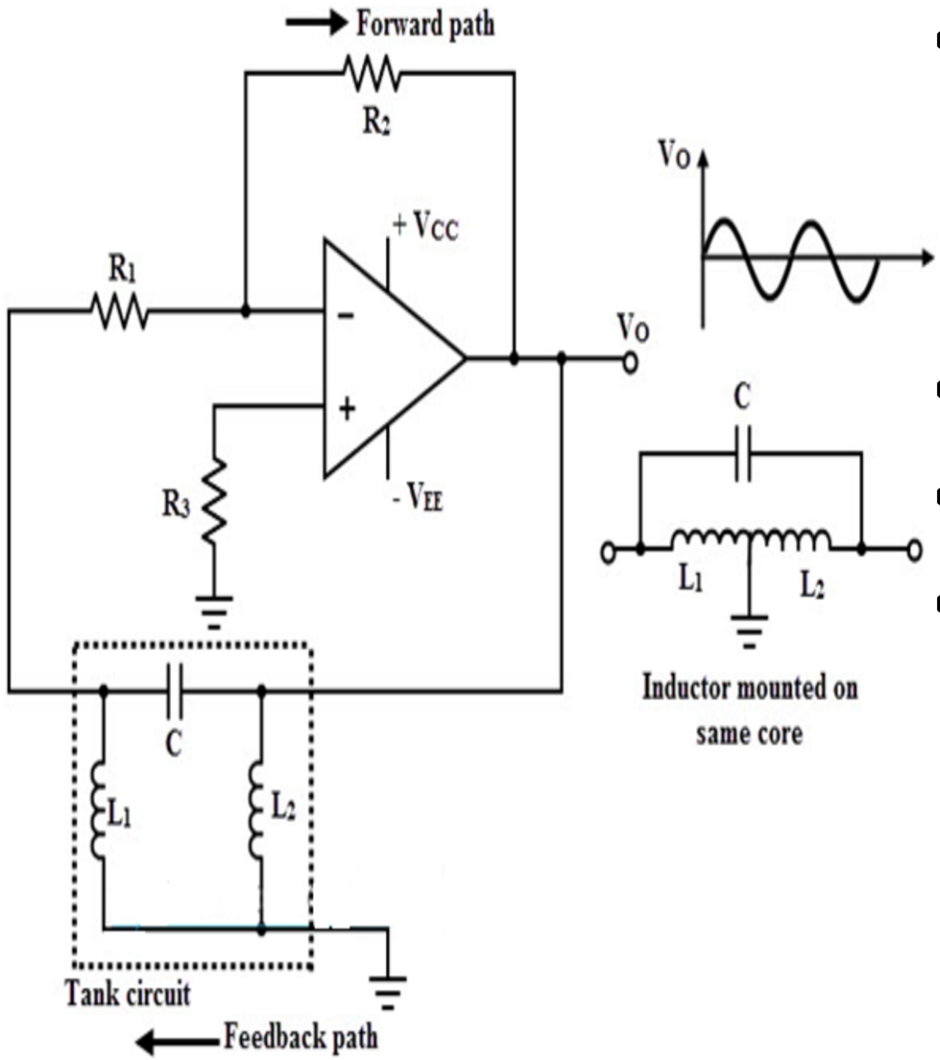
Hartley Oscillator



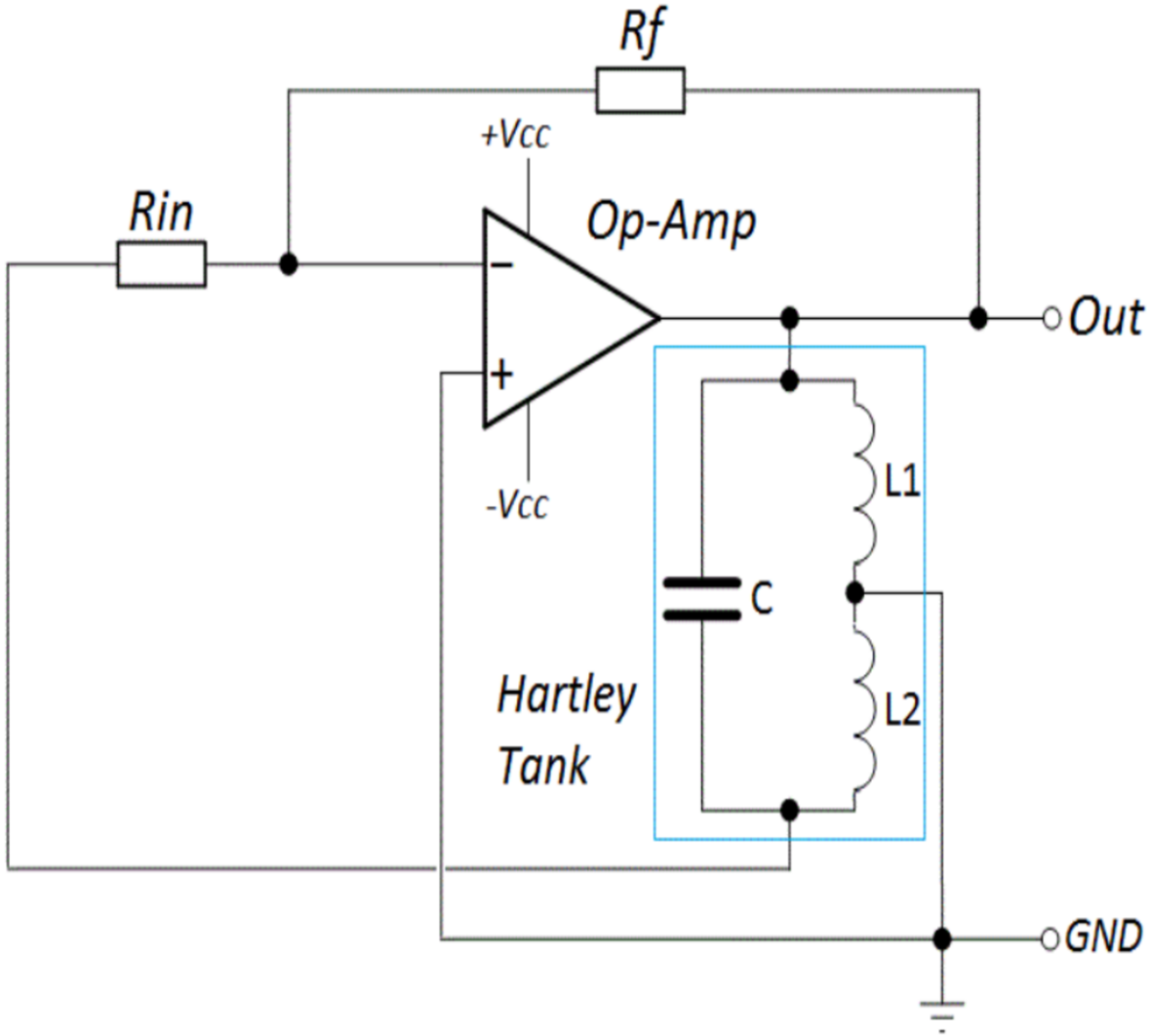
- The **Hartley oscillator** is one of the LC type (Inductor-Capacitor) feedback oscillator
- Gain adjustment using the feedback resistance and input resistance.
- The sine wave generated by the feedback circuit is coupled with the op-amp section. Then this wave is stabilized and inverted by the amplifier.

If the mutual inductance exists between L1 and L2 because of the common core for these two coils, then the gain becomes

$$A_V = \frac{L_1+M}{L_1+M}$$

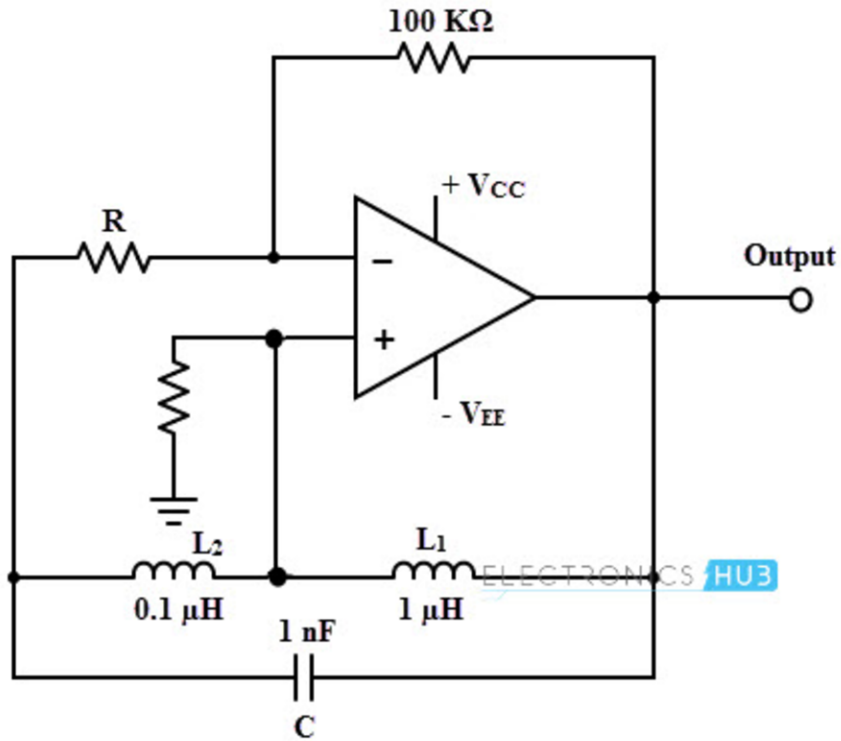


- The frequency of the oscillator is varied by using variable capacitor in the tank circuit keeping the feedback ratio and amplitude of the output constant for over a frequency range.
- The frequency of oscillations
- $Fo = 12\pi(Leq\ \ C)$ where $L_{eq} = L_1 + L_2$
- To generate the oscillation from this circuit, the amplifier gain must be selected greater than or at least equal to the ratio of two inductances. $Av = L_1\ /L_2$



The Op-amp is connected in an inverting configuration.
The resistor R1 and R2 is the feedback resistor.
OP amp gain is - R2/R1

- Consider the given figure, in which Hartley oscillator is constructed with operational amplifier and feedback LC network. By referring the given values determine the operating frequency and maximum acceptable value of resistance R for oscillations to start.



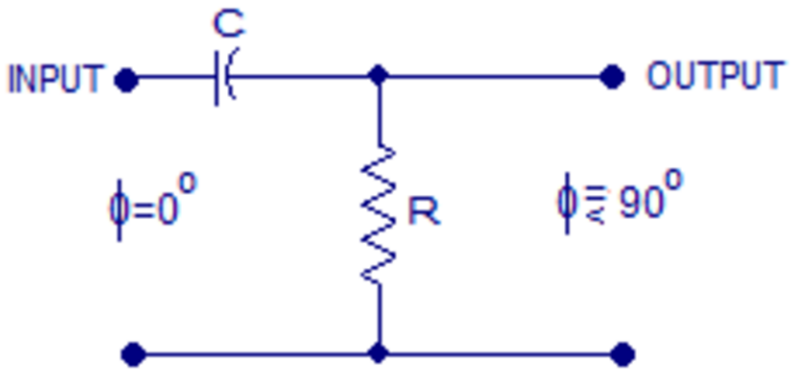
Source: www.electronicshub.org

Calculate

1. Fo = $\frac{1}{2\pi(Leq\ C)}$ where Leq = $L_1 + L_2$
2. Feedback factor = L2 / L1
3. Minimum gain = 1 / Feedback factor
4. Gain = R2 /R = Minimum gain
5. So calculate R

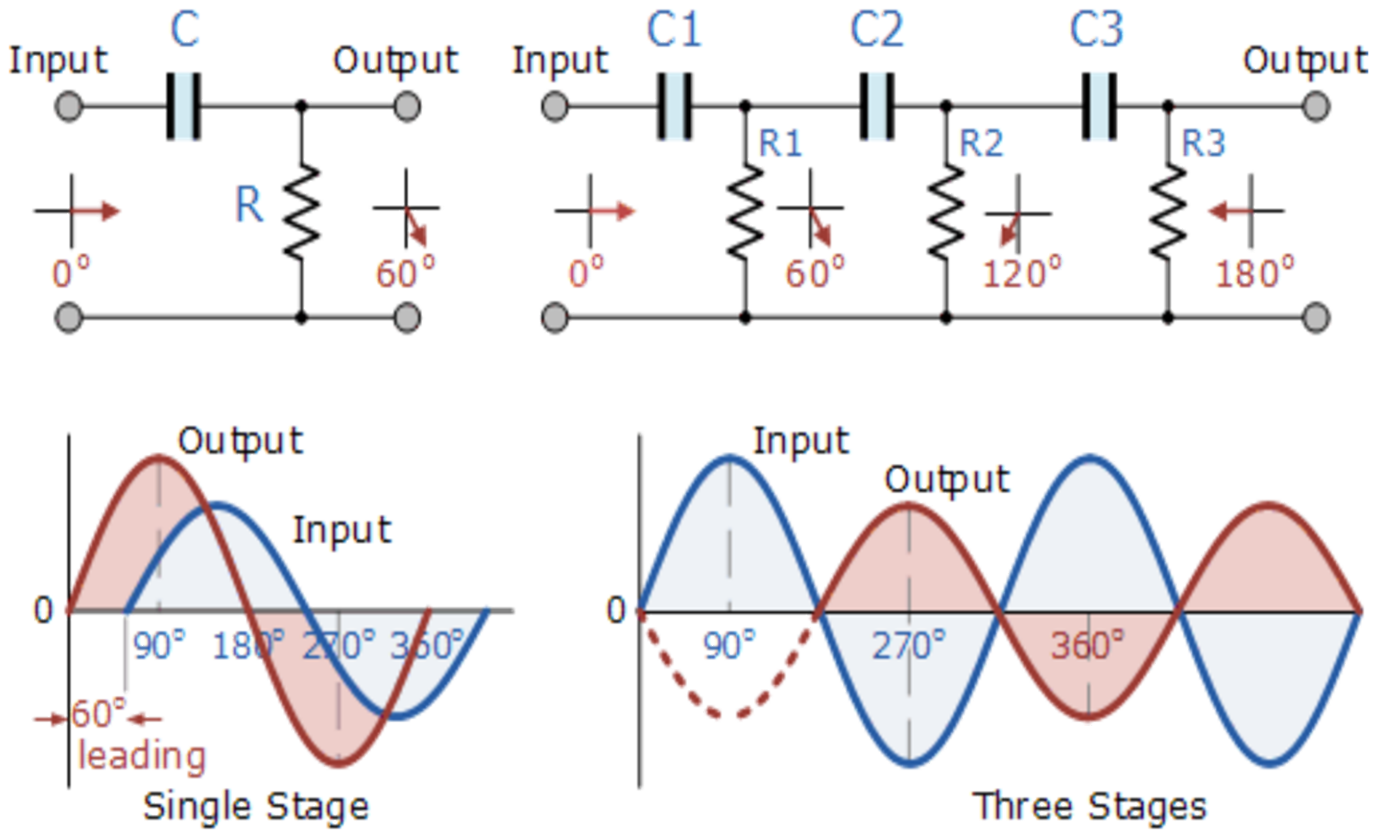
- **Advantages**
- Instead of two separate coils as L1 and L2, a single coil of bare wire can be used and the coil grounded at any desired point along it.
- By using variable capacitor or by making core movable (varying the inductance), frequency of oscillations can be varied.
- The amplitude of the output remains constant over the working frequency range.
- Very few components are needed including either two fixed inductors or a tapped coil.
- **Disadvantages**
- It cannot be used as low frequency oscillator since the value of inductors become large and size of the inductors becomes bulky.
- The harmonic content in the output of this oscillator is very high and hence it is not suitable for the applications which require pure sine wave.

RC phase shift oscillator



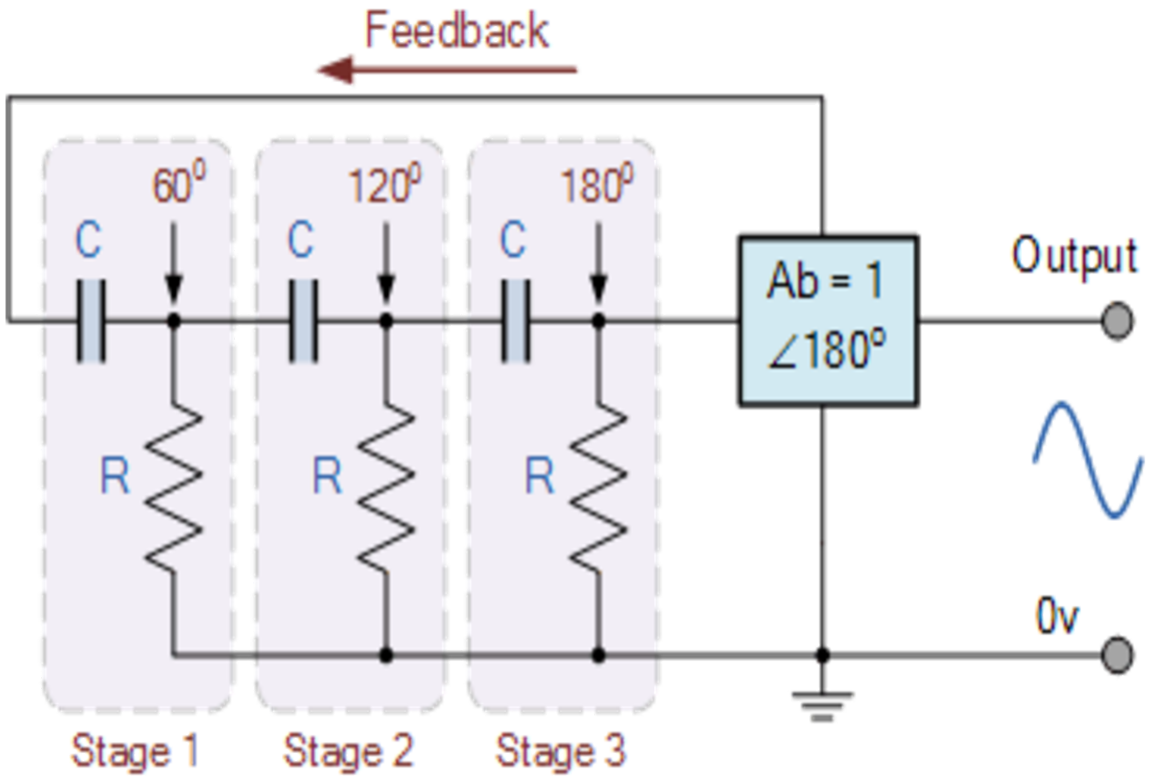
Single stage RC phase shift network

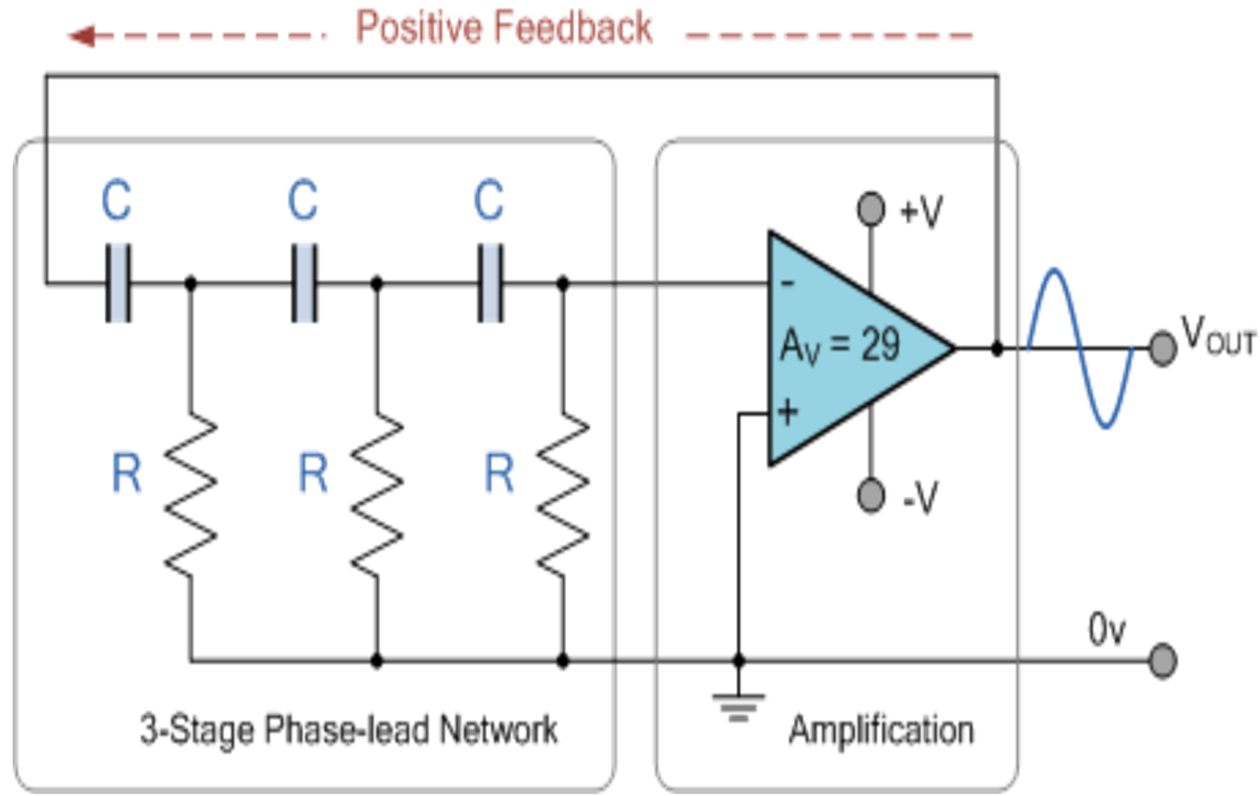
- RC oscillators are more suitable for low frequency applications.
- An RC oscillator is one of the sinusoidal oscillators and produces a sine wave output by using linear electronic components. The tuned LC oscillators operate well at higher frequencies, but at low frequencies, the capacitors and inductors in tank circuit or time circuit would be very large size.



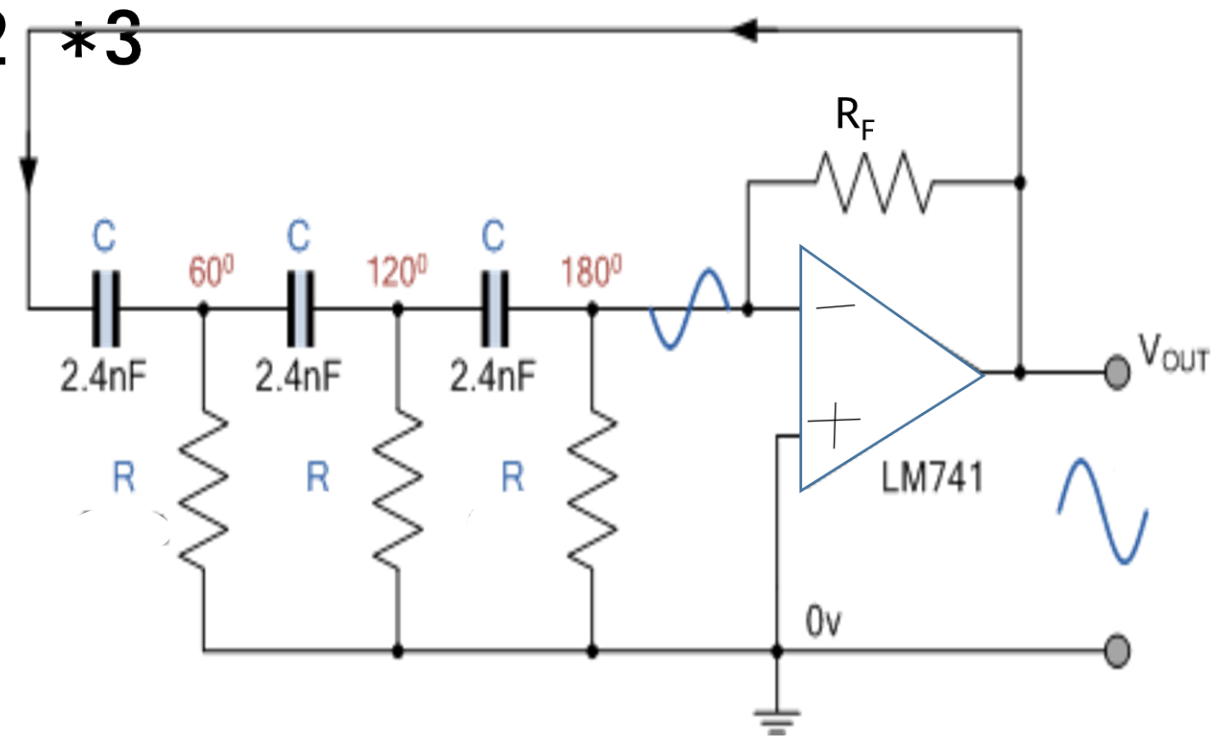
Source: https://www.electronics-tutorials.ws/

- The circuit on the left shows a single resistor-capacitor network whose outpu voltage “leads” the input voltage by some angle less than 90°.
- In a pure or ideal single-pole RC network. it would produce a maximum phase shift of exactly 90°, and because 180° of phase shift is required for oscillation, at least two single-poles networks must be used within an *RC oscillator* design.
- However in reality it is difficult to obtain exactly 90° of phase shift for each RC stage so we must therefore use more RC stages cascaded together to obtain the required value at the oscillation frequency.
- RC phase angle $\phi_c = \tan^{-1} \boldsymbol{XcR}$


Feedback
60°
120°
180°
C
C
C
R
R
R
Ab = 1
∠180°
Output
0v
Stage 1
Stage 2
Stage 3


Positive Feedback
C
C
C
R
R
R
+V
-V
AV = 29
VOUT
0v
3-Stage Phase-lead Network
Amplification

- An operational amplifier based *3-stage RC Phase Shift Oscillator* is required to produce a sinusoidal output frequency of 4kHz.
  If 2.4nF capacitors are used in the feedback circuit, calculate the value of the frequency determining resistors and the value of the feedback resistor required to sustain oscillations.
- F = $12\pi RC2N$ N = no of stages = 3
- 4 K = $12*\pi*R*2.4*10-9*2*3$
- R = ?
- Av = Rf/ R (Av should be 29)
- Rf = 29 * R

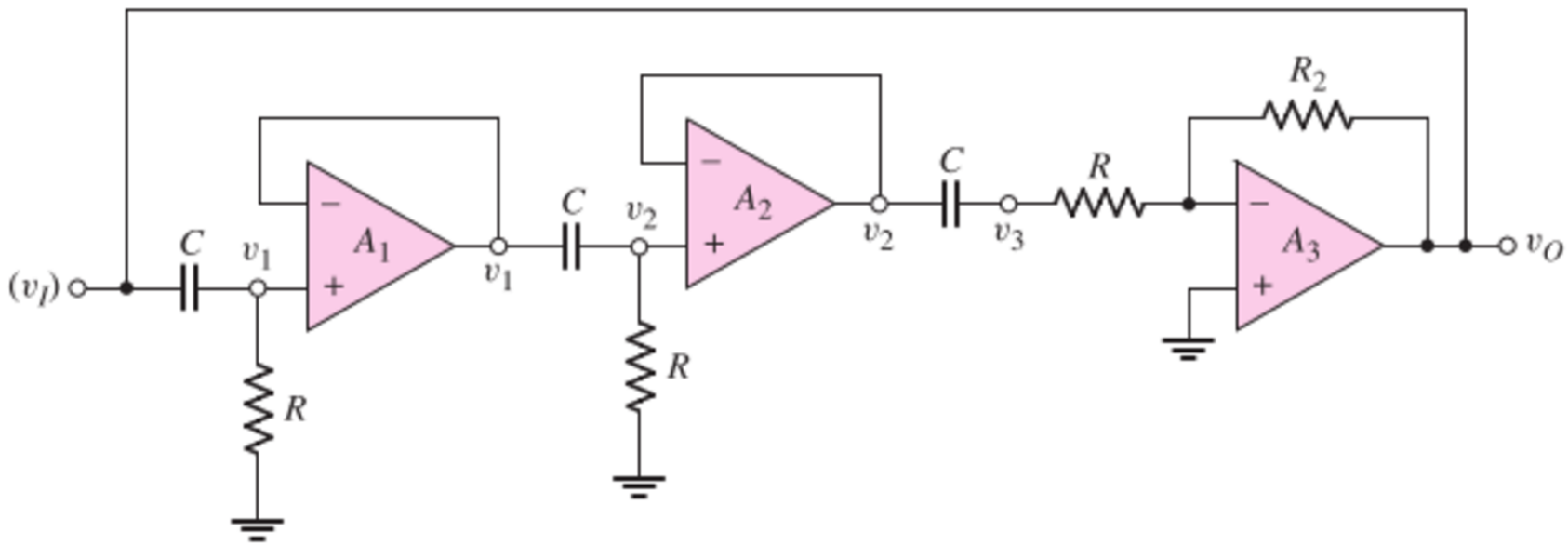


Advantages of Phase Shift Oscillators

- Due to the absence of expensive and bulky high-value inductors, circuit is simple to design and well suited for frequencies below 10 KHz.
- These can produce pure sinusoidal waveform since only one frequency can fulfill the Barkhausen phase shift requirement.
- It is fixed to one frequency.

Disadvantages of Phase Shift Oscillators

- For a variable frequency usage, phase shift oscillators are not suited because the capacitor values will have to be varied. And also, for frequency change in every time requires gain adjustment for satisfying the condition of oscillations.
- These oscillators produce 5% of distortion level in the output.
- This oscillator gives only a small output due to smaller feedback
- These oscillator circuits require a high gain which is practically impossible.
- The frequency stability is poor due to the effect of temperature, aging, etc. of various circuit components.



$$\omega_o = \frac{1}{\sqrt{3}RC}$$

$$\frac{R_2}{R} = 8$$

Figure 15.16 Phase-shift oscillator circuit with voltage-follower buffer stages

Basic Amplifier is A3 connected as inverting amplifier.

Voltage follower eliminate loading effect between each RC filter stage

$$A(s) = \left|\frac{v_O}{v_3}\right| = \frac{R_2}{R}$$

The loop gain is then

$$T(s) = A(s)\beta(s) = \left(\frac{R_2}{R}\right)\left(\frac{sRC}{1+sRC}\right)^3$$

EXAMPLE 15.4

Objective: Determine the oscillation frequency and required amplifier gain for a phase-shift oscillator.

Consider the phase-shift oscillator in Figure 15.16 with parameters $C = 0.1\ \mu\text{F}$ and $R = 1\ \text{k}\Omega$.

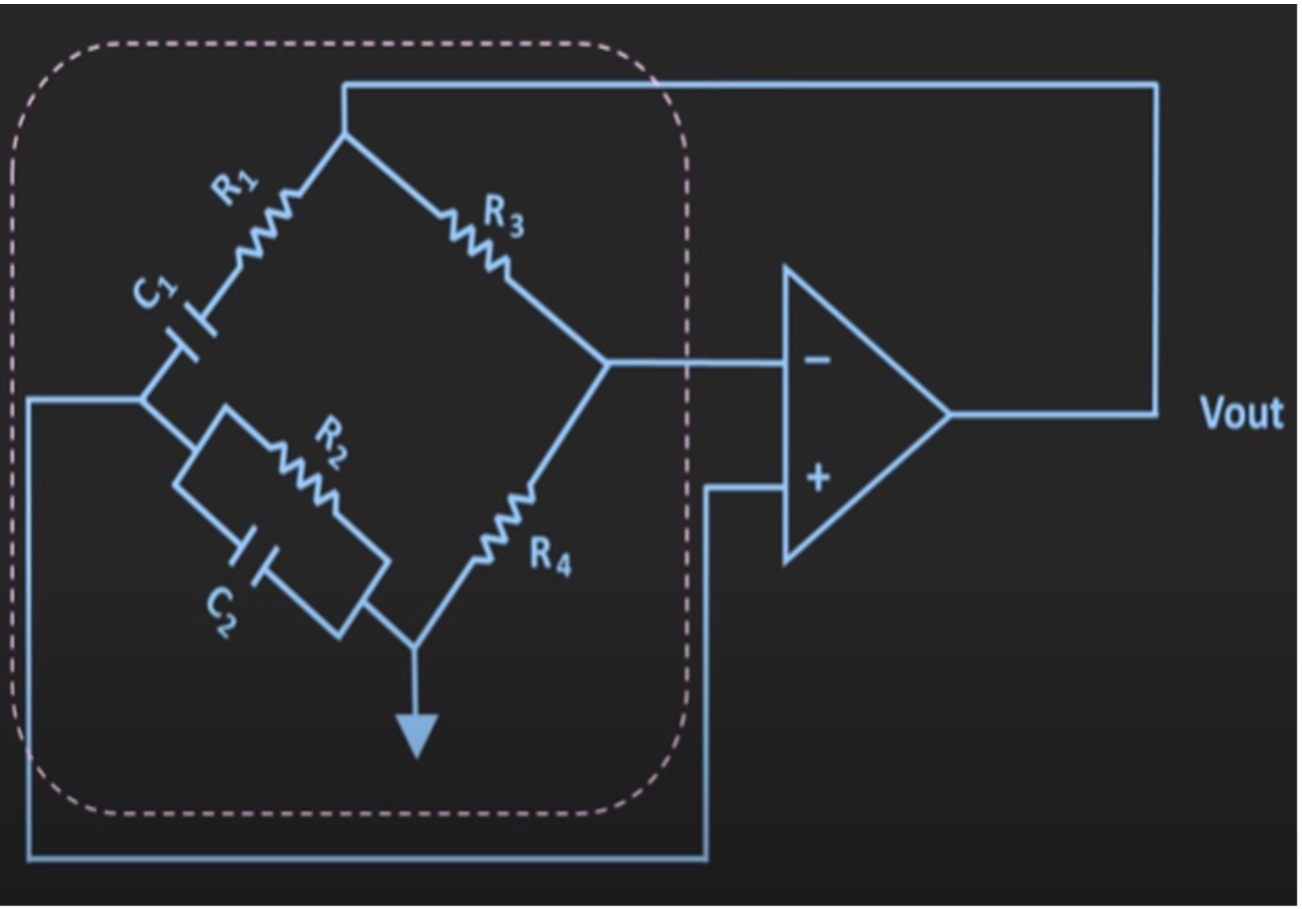
Solution: From Equation (15.41), the oscillation frequency is

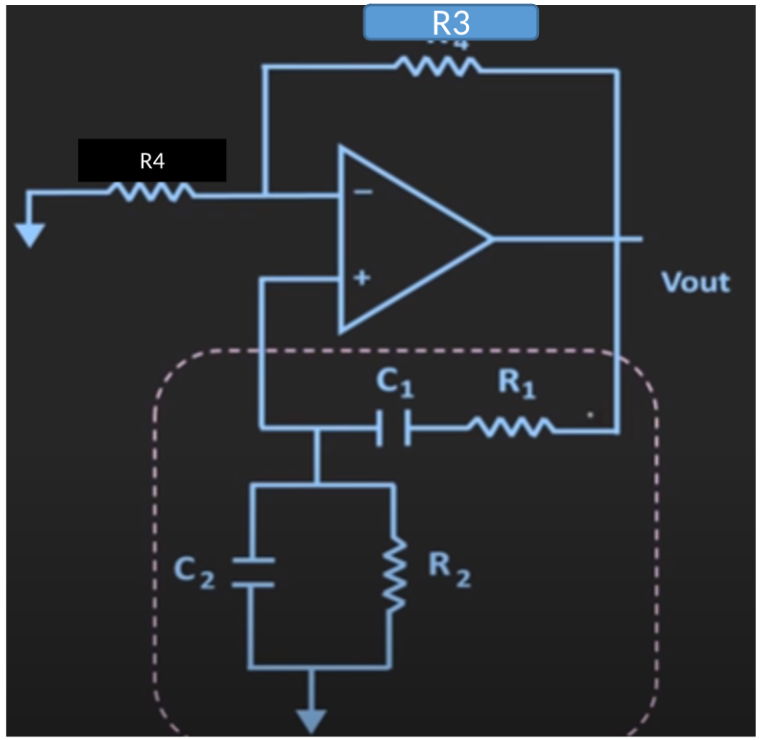
$$f_o = \frac{1}{2\pi\sqrt{3}RC} = \frac{1}{2\pi\sqrt{3}(10^3)(0.1 \times 10^{-6})} = 919\,\text{Hz}$$

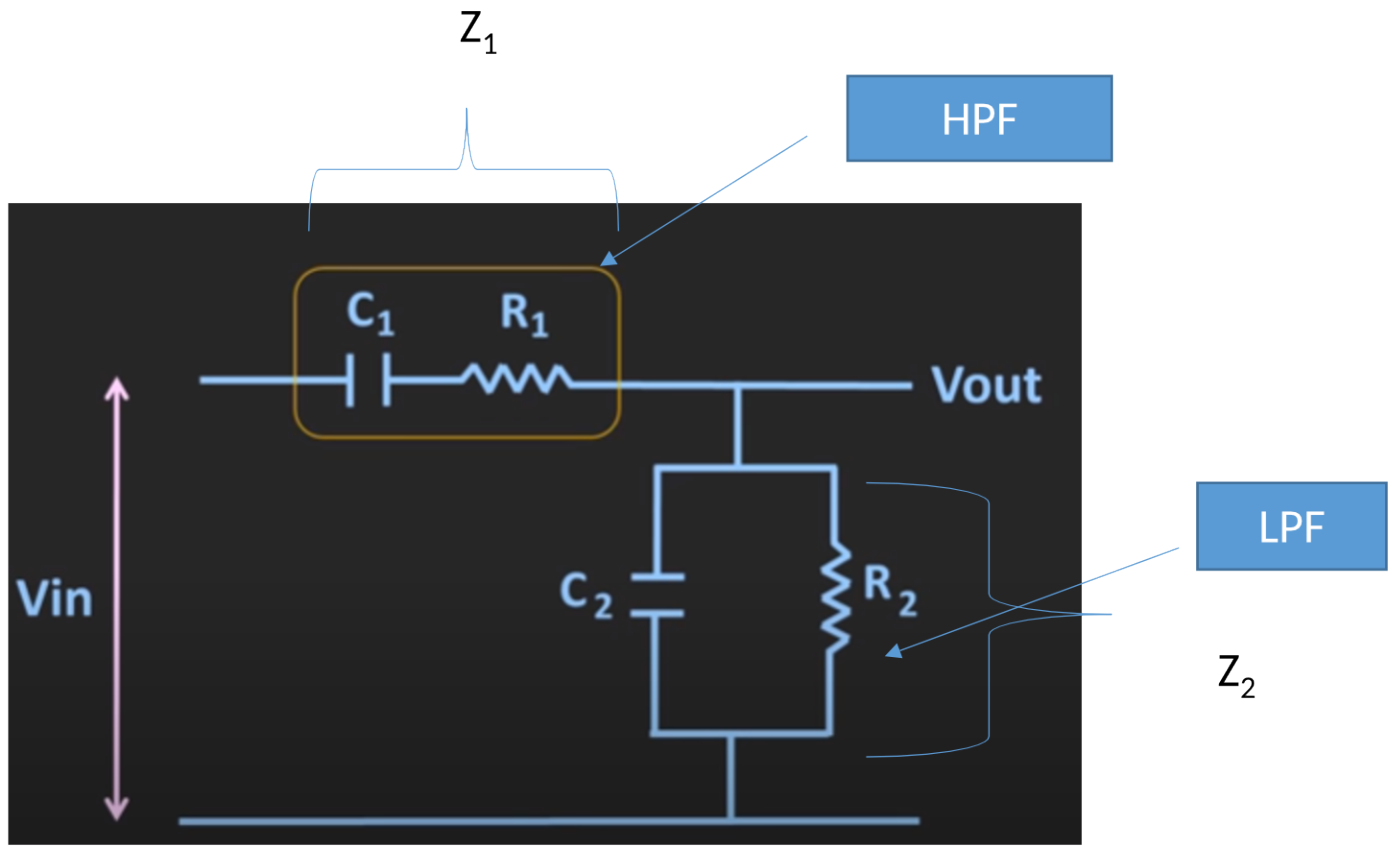
The minimum amplifier gain magnitude is 8 from Equation 15.43; therefore, the minimum value of R_2 is $8\ \text{k}\Omega$.

Comment: Higher oscillation frequencies can easily be obtained by using smaller capacitor values.

Wein Bridge Oscillator







Acts like notch filter

Vo/Vin = 1/3 = feedback factor

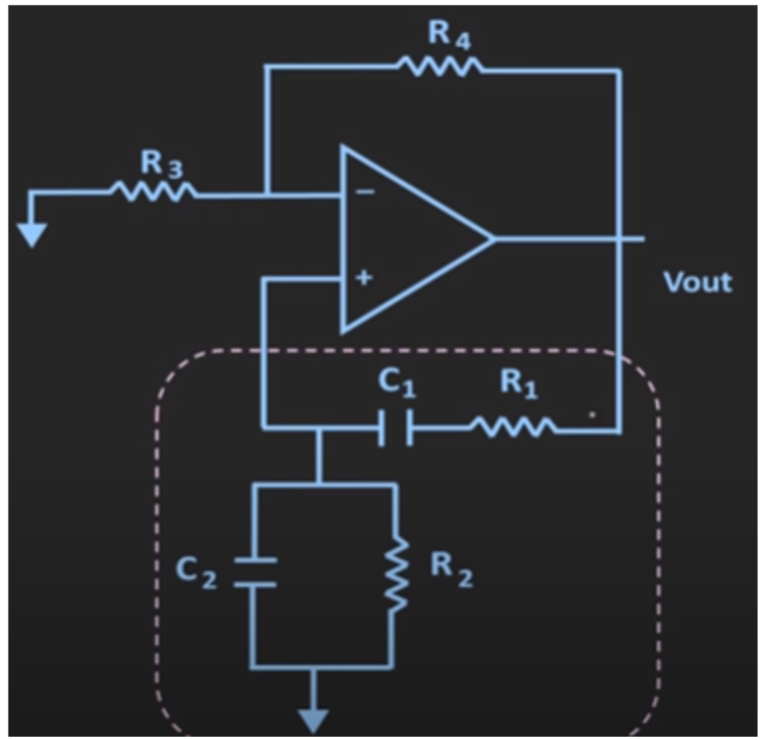
Resonant frequency F = $12\pi RC$

if R1 = R2

and C1 = C2

Otherwise F = $12\pi\ R1R2C1C2$

$VoutVin = Z2\ Z1+Z2$



Feedback factor $\beta = 1/3$

To get sustained oscillation $A\beta = 1$

$A = 3$

Gain $A = 1 + \frac{R4}{R3} = 3$

So, R4 = 2 R3

$$\frac{Vout}{Vin} = \frac{Z2}{Z1+Z2}$$

$Z_2 = R_2 \,//\, (1j\omega C2)$

$Z_1 = R_1 + (1j\omega C1)$

- $\frac{Vout}{Vin} = \frac{j\omega R2C1}{1 - \omega^2 R1R2C1C2 + j\omega(R1C1 + R2C2 + R2C1)}$
- At resonant frequency phase shift offered should be zero.
- $(1 - \omega^2 R1R2C1C2) = 0$; so that j ω from numerator and denominator will get cancelled to provide zero phase shift.

$\omega^2 R1R2C1C2 = 1$ $\quad F = \frac{1}{2\pi} \frac{1}{R1R2C1C2}$ $\quad \beta = 1/3$

$$\frac{Vout}{Vin} = \frac{R2C1}{R1C1 + R2C2 + R2C1} = \beta$$

$$A = 1/\beta = \frac{R1C1 + R2C2 + R2C1}{R2C1} = \frac{R1}{R2} + \frac{C2}{C1} + 1$$

$1 + \frac{R4}{R3} = \frac{R1}{R2} + \frac{C2}{C1} + 1$ $\quad \frac{R4}{R3} = \frac{R1}{R2} + \frac{C2}{C1}$ $\quad$ So, $F = \frac{1}{2\pi RC}$;

and C1 = C2



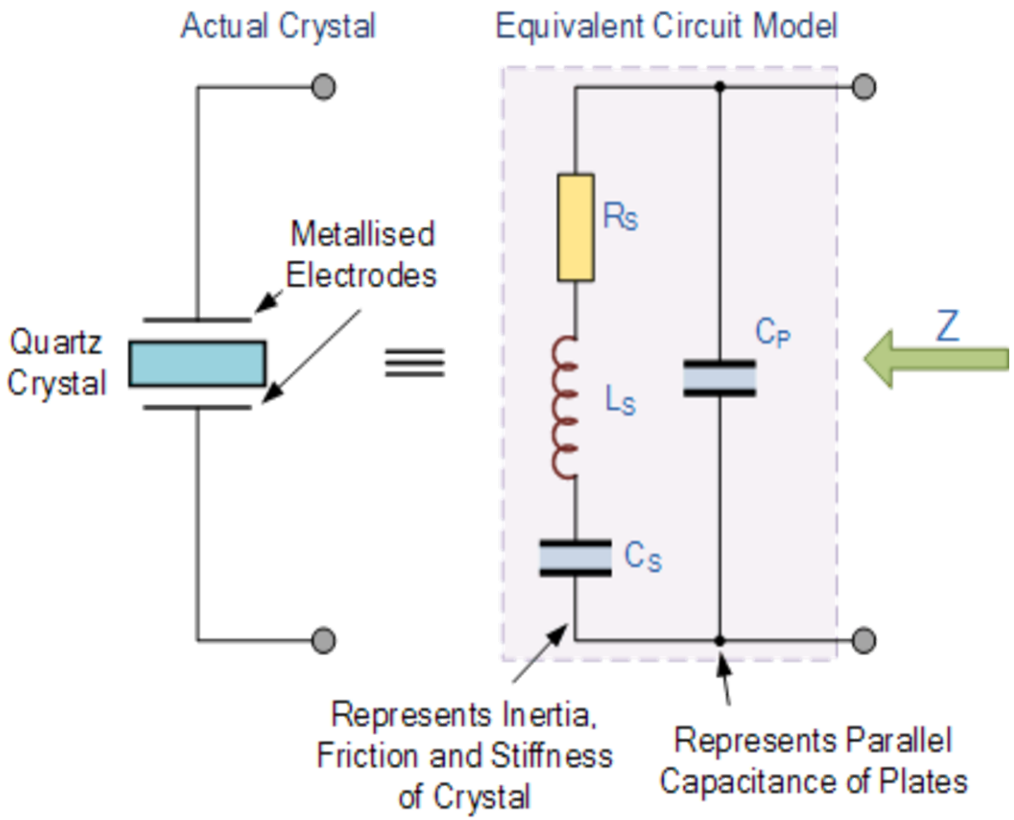
Crystal Oscillator

- A **crystal oscillator** is an electronic oscillator circuit that uses the mechanical resonance of a vibrating crystal of piezoelectric material to create an electrical signal with a precise frequency.
- A crystal oscillator, particularly one using a quartz crystal, works by distorting the crystal with an electric field, when voltage is applied to an electrode near or on the crystal; a property known as electrostriction or inverse piezoelectricity.
- When the electric field is removed, the quartz—which oscillates at a precise frequency—generates an electric field as it returns to its previous shape, and this can generate a voltage.
- The result is that a quartz crystal behaves like an RLC circuit, but with a much higher Q.

- In crystal oscillators, the usual electrical resonant circuit is replaced by a mechanically vibrating crystal.
- The crystal (usually quartz) has a high degree of stability in holding constant at whatever frequency the crystal is originally cut to operate.
- The crystal oscillators are, therefore, used whenever great stability is needed, for example, in communication transmitters, and receivers, digital clocks etc.
- A quartz crystal exhibits a very important property known as piezo-electric effect. When a mechanical pressure is applied across the faces of the crystal, a voltage proportional to the applied mechanical pressure appears across the crystal. Conversely, when a voltage is applied across the crystal surfaces, the crystal is distorted by an amount proportional to the applied voltage. An alternating voltage applied to a crystal causes it to vibrate at its natural frequency.



Quartz Crystal Equivalent Model



Series Resonant frequency

$$f_S = \frac{1}{2\pi\sqrt{L_S C_S}}$$

Parallel Resonant frequency

$$f_P = \frac{1}{2\pi\sqrt{L_S\left(\frac{C_P C_S}{C_P + C_S}\right)}}$$

- The equivalent electrical circuit for the quartz crystal shows a series RLC circuit, which represents the mechanical vibrations of the crystal, in parallel with a capacitance, Cp which represents the electrical connections to the crystal. Quartz crystal oscillators tend to operate towards their “series resonance”.
- The equivalent impedance of the crystal has a series resonance where Cs resonates with inductance, Ls at the crystals operating frequency. This frequency is called the crystals series frequency, fs. As well as this series frequency, there is a second frequency point established as a result of the parallel resonance created when Ls and Cs resonates with the parallel capacitor Cp as shown.

- *Crystal oscillator circuits* are generally constructed using bipolar transistors or FETs. This is because although operational amplifiers can be used in many different low frequency (≤100kHz) oscillator circuits, operational amplifiers just do not have the bandwidth to operate successfully at the higher frequencies suited to crystals above 1MHz.

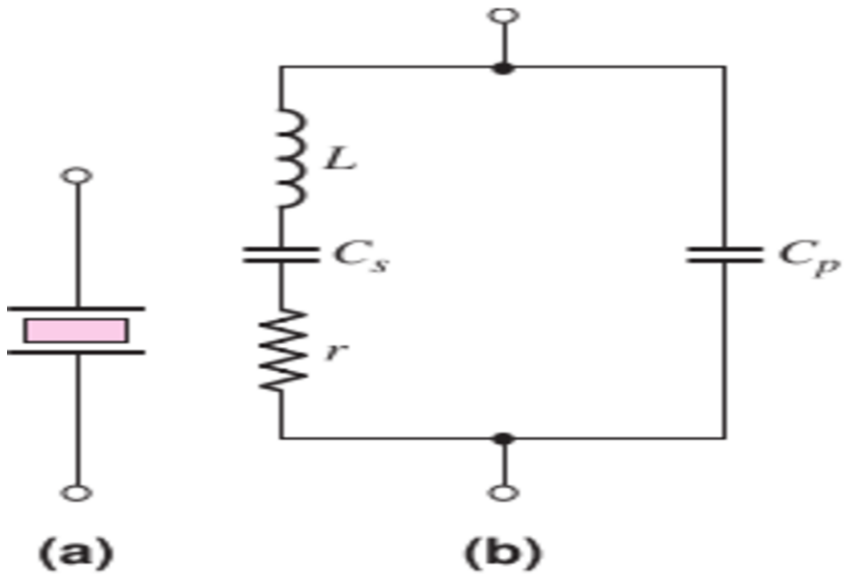
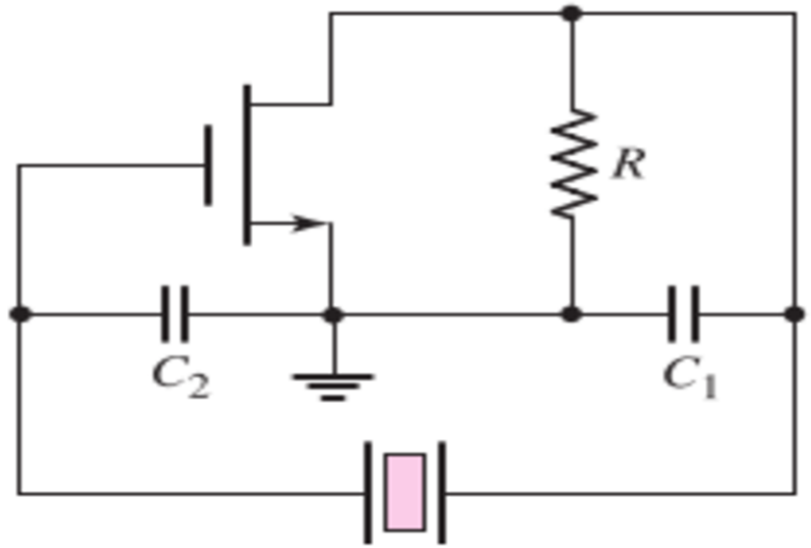


Figure 15.23 (a) Piezoelectric crystal circuit symbol and (b) piezoelectric crystal equivalent circuit



Reference

- Electronic circuit analysis by Donald Neaman
- https://www.electronicshub.org/hartley-oscillator/
- https://circuitdigest.com/tutorial/hartley-oscillator
- https://www.youtube.com/watch?v=SrvkXEIWuK4